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Enhancing Offshore Field Development Optimization Through AI Asset Collaboration Excellence (ACE)

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Abstract

This document provides a comprehensive analysis of ACE to improve the quality and speed of decision making, driven by data insights, enriching the collaboration between teams and individuals, which results in tangible business benefits being achieved. Through an in-depth examination of ACE adoption and its gradual maturity along with AI, it is expected that the need for direct human intervention will be reduced in future while ensuring that the operations are managed with precision and agility, adapting to real-time data and evolving conditions. Ultimately, the goal is to create a seamless, intelligent operational environment that drives innovation and maximizes value for all stakeholders.

The findings present underscore the critical importance of A key pillar in this journey is the establishment of an AI-enabled Asset Collaboration Excellence ACE harnessing synergies of people, process, and technology. The document aims to offer valuable insights and recommendations for ADNOC. By leveraging advanced technologies and integrating smarter, collaborative decision-making frameworks, it seeks to achieve enhance operational efficiency, safety, and sustainability. This abstract serves as a concise summary of the document's key points and overarching message, providing readers with a clear understanding of its content and purpose.

Introduction

ADNOC has set a long-term objective of achieving autonomous operations, which is centered on the continuous optimization of all their resources through automated, data-driven processes. This ambitious goal is driven by the need to enhance efficiency, reduce operational costs, and improve overall productivity. A key pillar in this journey is the establishment of an AI-enabled Collaborative Work Environment (ACE), harnessing synergies of people, processes, and technology. The ACE aims to create a seamless and integrated workspace where stakeholders can collaborate effectively, leveraging advanced technologies to drive innovation and achieve strategic objectives.

In that context, a project (ACE Conceptual Design) was launched with the objective of defining the level 01 (connecting the Site to L2) of the AOF collaboration strategy. The project methodology included the phases of Assess-Analyze-Design. The Assess phase involved comprehensive information gathering through stakeholder interviews and workshops. This phase was crucial in understanding the current state of

operations, identifying gaps, and gathering insights from various stakeholders. Findings and requirements for collaboration and ACE design were meticulously compiled and structured, ensuring that the design would address the specific needs and challenges faced by the organization.

Based on the analysis of the findings, ACE operating models and governance were defined. Out of the multiple options explored for the ACE operating models, each analyzed on their own pros and cons and based on management guidance, a model with ACE for assets grouped as per their common flow stream is proposed. This approach ensures that the ACE is tailored to the unique operational characteristics of each asset group, promoting more efficient and effective collaboration.

An initial ACE conceptual design was created. After syndication with stakeholders, the design was modified to produce an "Optimized" design considered by removing certain features which were deemed to make the ACE more "Fit-for-purpose". This iterative process of refining the design based on stakeholder feedback highlights the collaborative nature of the project and the commitment to creating a solution that meets the needs of all users. After due consideration, there was a pivot to the virtual ACE concept and possibly a hybrid collaboration system (Physical and Virtual). This shift reflects the evolving nature of work environments and the need to accommodate remote and flexible working arrangements. With this approach, the Optimized ACE design was scaled to a "Minimized" design, focusing on the most critical features and functionalities.

To better understand virtual ACE, a workshop was held to start defining and conceptualizing virtual ACE. Virtual ACE was defined, and several potential technologies to enable virtual ACE were identified. This exploration of virtual collaboration tools and technologies is essential in ensuring that the ACE remains adaptable and future-proof, capable of supporting ADNOC's long-term objectives of autonomous operations and continuous optimization.

A comprehensive business case and associated value estimation were created to outline the potential benefits and justify the investment in the proposed project. The value sources were meticulously identified and included the following key areas:

- **Improvement in Production:** By implementing advanced technologies and optimizing processes, the project aims to enhance overall production efficiency. This includes reducing downtime, increasing throughput, and improving the quality of the output. The anticipated improvements in production will not only boost the overall output but also contribute to higher revenue generation and better resource utilization.
- **Reduction in Planned Deferment:** Planned deferments, which are scheduled interruptions in production for maintenance or other activities, can be minimized through better planning and execution. The project will introduce predictive maintenance techniques and advanced scheduling tools to ensure that planned deferments are carried out more efficiently, thereby reducing their frequency and duration. This will lead to more consistent production levels and reduced operational disruptions.
- **Reduction in Unplanned Deferment:** Unplanned deferments, caused by unexpected equipment failures or other unforeseen issues, can have a significant impact on production. The project will leverage real-time monitoring and predictive analytics to identify potential issues before they lead to unplanned deferments. By addressing these issues proactively, the project aims to minimize the occurrence of unplanned deferments, ensuring smoother and more reliable operations.
- **Improvement in Equipment Availability:** Ensuring that equipment is available and operational when needed is crucial for maintaining high production levels. The project will implement advanced maintenance strategies, such as condition-based monitoring and predictive maintenance, to improve equipment availability. By reducing the frequency and duration of equipment downtime, the project will enhance overall operational efficiency and productivity.

- **Reduction in Maintenance Backlog:** A significant maintenance backlog can lead to operational inefficiencies and increased risk of equipment failure. The project will introduce streamlined maintenance processes and better resource allocation to address the maintenance backlog more effectively. By reducing the backlog, the project will ensure that maintenance activities are carried out in a timely manner, contributing to improved equipment reliability and overall operational performance.

ACE Definition

ACE is defined as a workspace where business processes are managed, and decisions are made, enabled by collaborative communication, using shared data with common visualization. It provides seamless communication between disciplines and between teams in various locations. With the ultimate business objective for ADNOC Offshore being the reduction of deferments and achieving production targets in a cost-effective manner while adhering to HSE constraints, other goals of ADNOC Offshore's new ACE design are:

- **Enhance Communications and Team Integration:** Effective communication and team integration are crucial for the success of any organization. The ACE design aims to break down communication barriers and foster a collaborative environment where team members can easily share information, ideas, and feedback. This will lead to better decision-making, increased efficiency, and a more cohesive team.
- **Improve Business Performance through Collaborative Data-Driven Decision Making:** In today's data-driven world, making informed decisions based on accurate and timely data is essential. The ACE design will enable collaborative data-driven decision-making by providing a platform where data from various sources can be integrated, analyzed, and visualized. This will help teams make better decisions, optimize processes, and improve overall business performance.
- **Boost Productivity via Accelerated Process Execution and Ensuring Situational Awareness:** The ACE design aims to streamline processes and reduce the time required to execute tasks. By providing real-time situational awareness and access to relevant information, team members can quickly respond to changing conditions and make informed decisions. This will lead to increased productivity and more efficient operations.
- **Foster a Culture of Teamwork and Cross-Functional Collaboration, Eliminating Silos:** Silos can hinder collaboration and innovation. The ACE design aims to create a culture of teamwork and cross-functional collaboration by encouraging open communication and the sharing of knowledge and resources. This will help eliminate silos, promote innovation, and drive organizational success.
- **Increase Employee Job Satisfaction:** A collaborative and supportive work environment can significantly impact employee job satisfaction. The ACE design aims to create a positive work environment where employees feel valued, supported, and motivated. This will lead to higher job satisfaction, increased employee retention, and a more engaged workforce.
- **Improve the Competency of Asset Staff Responsible for the Business Operations Concerned:** Continuous learning and development are essential for maintaining a skilled and competent workforce. The ACE design will provide opportunities for training, knowledge sharing, and skill development, helping asset staff stay up to date with the latest industry trends and best practices. This will improve their competence and contribute to the overall success of the organization.
- **Accelerate Digital Transformation Including the Adoption of AI Technologies:** Digital transformation is key to staying competitive in today's fast-paced business environment. The ACE design will support the adoption of advanced technologies, including AI, to automate processes, enhance decision-making, and drive innovation. This will help ADNOC Offshore stay ahead of the curve and achieve its strategic objectives.

- **Create Enablers for Future Autonomous and Remote Process Execution:** The ACE design aims to lay the foundation for future autonomous and remote process execution. By leveraging advanced technologies and creating a collaborative work environment, ADNOC Offshore will be well-positioned to implement autonomous operations and remote work, leading to increased efficiency, reduced costs, and improved operational performance.

ACE concept development

Following the historical and business context and the project methodology (Assess/Analyze/Design) of the previous section, this section describes of ACE design requirements were formulated and describe. Requirements are anchored in the AOF Collaboration vision and strategy described below.

The project team was tasked with proposing a conceptual design for a standardized Collaborative Work Environment (ACE) that would enable decision-making and insights as a key capability to support SMART objectives. This ACE is envisioned to play an important role in ADNOC's Digital Transformation roadmap, aiming to achieve collaboration goals across nine offshore assets and two terminals. The standardized design also needs to be adapted to one of the selected assets, ensuring it meets the specific needs and challenges of that asset.

A three-tiered approach was adopted to deliver the conceptual design of the ACE, which was based on a proven methodology, industry best practices, and prior ACE design engagements. This approach ensured a comprehensive and systematic process for developing the ACE design, addressing all critical aspects of collaboration and decision-making.

The first tier focused on **Assessing** the current state of operations and identifying the specific needs and challenges of the organization. This involved comprehensive information gathering through stakeholder interviews, workshops, and site visits. The goal was to understand the existing processes, communication channels, and data management practices, as well as to identify any gaps or areas for improvement.

The second tier involved **Analyzing** the gathered information to develop a clear understanding of the requirements for the ACE. This phase included evaluating different ACE operating models, assessing their pros and cons, and determining the most suitable model based on management guidance and organizational needs. The analysis also considered the governance structure required to support the ACE and ensure its effective implementation and operation.

The third tier focused on **Designing** the conceptual ACE, incorporating the insights and findings from the assessment and analysis phases. An initial ACE conceptual design was created, which was then refined through syndication with stakeholders. This iterative process ensured that the design was optimized to be fit-for-purpose, addressing the specific needs and challenges identified earlier. The design was further scaled to a minimized version, focusing on the most critical features and functionalities, and considering the potential for a hybrid collaboration system (physical and virtual).

Virtual ACE Concept

The project also explored the concept of a virtual Collaborative Work Environment (ACE), holding workshops to define and conceptualize this approach. These workshops brought together experts from various disciplines to brainstorm and discuss the requirements and potential benefits of a virtual ACE. The collaborative nature of these workshops ensured that diverse perspectives were considered, leading to a well-rounded and comprehensive understanding of the virtual ACE concept.

Several potential technologies to enable virtual ACE were identified during these workshops. These technologies include advanced communication tools, real-time data visualization platforms, and AI-driven analytics solutions. By leveraging these technologies, virtual ACE can provide seamless communication and collaboration between team members, regardless of their physical location. This is particularly important in today's increasingly remote and distributed work environments.

The virtual ACE is designed to be adaptable and future-proof, ensuring that it can evolve with the changing needs of the organization. This adaptability is achieved using scalable technologies and modular design principles, allowing for easy integration of new tools and features as they become available. By staying at the forefront of technological advancements, virtual ACE can continue to support ADNOC's long-term objectives of autonomous operations and continuous optimization.

This exploration of virtual collaboration tools and technologies is essential in supporting ADNOC's vision of a fully integrated and efficient work environment. By embracing digital transformation and leveraging the latest technologies, ADNOC can enhance its operational capabilities, improve decision-making processes, and drive innovation across the organization. Virtual ACE represents a significant step towards achieving these goals, providing a robust and flexible platform for collaboration and decision-making.

Findings and Requirements

Some of the major findings based on insights generated from interviews, workshops, and other discussions, combined with industry best practices and the project team's experience, are summarized below:

People:

- **Strong Governance Rules and Guidelines:** Ensuring operational effectiveness from asset to ACE requires robust governance rules and guidelines. These rules provide a clear framework for decision-making, accountability, and performance measurement. By establishing strong governance, the organization can ensure that all team members are aligned with strategic objectives and that operations are conducted efficiently and transparently.
- **Co-located Teams Aligned Towards Shared Goals and Common KPIs:** Co-located teams, working towards shared goals and common Key Performance Indicators (KPIs), can collaborate more effectively. The ability to quickly access and collaborate with step-in/step-out and remote roles enhances flexibility and responsiveness. This alignment fosters a sense of unity and purpose, driving better performance and achieving desired outcomes.
- **Clear Change Management Principles:** Effective change management is crucial for the successful implementation of the ACE. Clear principles enable open communication, ensuring that all personnel are well-informed and trained for ACE operations. A phased implementation approach with incremental changes helps manage the transition smoothly, reducing resistance and ensuring that the organization adapts effectively to new processes and technologies.

Process:

- **AI-Enabled Work Processes:** Integrating AI into work processes can significantly enhance efficiency and decision-making. AI can automate routine tasks, analyze large datasets to provide actionable insights, and support predictive maintenance and optimization. This leads to more informed decisions, reduced operational costs, and improved productivity.
- **Tight Integration to L0 (Site) and L2 (Support Functions):** Ensuring tight integration between the site (L0) and support functions (L2) is essential for seamless operations. Improved integration with terminals and support functions ensures that information flows smoothly across all levels of the organization. This integration enhances coordination, reduces delays, and ensures that all parts of the organization are working towards common goals.
- **Standardized ACE Design:** A standardized ACE design provides the flexibility to scale up and evolve as business priorities shift. This design minimizes support costs by creating a consistent framework that can be easily adapted to different assets and operational contexts. Standardization ensures that best practices are consistently applied, leading to more efficient and effective operations.

Technology:

- **AI-Assisted/Enabled Decision Making:** The integration of AI into decision-making processes allows for more accurate and timely insights. AI can analyze vast amounts of data, identify patterns, and provide recommendations, enabling teams to make informed decisions quickly. This enhances the overall efficiency and effectiveness of operations, leading to better outcomes and reduced operational risks.
- **Leveraging Latest Technology and Integrating Disparate Applications:** By leveraging the latest technology and integrating disparate applications, the project aims to create a unified view with real-time data and metrics. This integration ensures that all relevant information is accessible in one place, providing a comprehensive overview of operations. Real-time data and metrics enable proactive decision-making, allowing teams to respond swiftly to changing conditions and optimize performance.
- **Advanced Communication Tools to Enable Smart Collaboration:** Advanced communication tools are essential for facilitating smart collaboration among team members. These tools enable seamless communication, whether through video conferencing, instant messaging, or collaborative platforms. By providing a robust communication infrastructure, the project ensures that teams can collaborate effectively, share information, and work together towards common goals.

Working Environment:

- **Design that Encourages Innovative Thought Process:** The design of the working environment plays a crucial role in fostering innovation. By creating a space that encourages creative thinking while ensuring safety, security, and a distraction-free environment, the project aims to enhance employee productivity. A well-designed workspace can inspire new ideas, promote collaboration, and drive innovation.
- **Essential Facilities Including AI-Enabled Collaboration Space, AR/VR Room, and Break Room:** Providing essential facilities such as AI-enabled collaboration spaces, AR/VR rooms, and break rooms is vital for supporting various work activities. AI-enabled collaboration spaces facilitate data-driven decision-making and teamwork, while AR/VR rooms offer immersive experiences for training, simulations, and design reviews. Break rooms provide a space for relaxation and informal interactions, contributing to employee well-being and productivity.
- **Modern, Ergonomic, Sustainable, and Aesthetically Designed:** A modern, ergonomic, sustainable, and aesthetically designed workspace enhances the overall work experience. Ergonomic design ensures that the workspace is comfortable and reduces the risk of work-related injuries. Sustainability initiatives demonstrate a commitment to environmental responsibility, while aesthetic design creates a pleasant and inspiring environment. Together, these elements contribute to a positive work culture and higher employee satisfaction.

Change management

The Collaborative Work Environment (ACE) will enable a new way of working, and like any other change, it's imperative to ensure that the effects of changes due to technology, organizational structure, and new ways of collaboration are well managed, and people are able to adapt to it. The success of the ACE hinges on its acceptance and use by its intended users. If people are resistant to change or unwilling to adopt new technologies, it will become difficult for the project to be successful. The full benefits can only be realized if users fully engage with the new tools and processes, which will require a shift in behavior and mindset.

Change Management Strategies.

- **Communication and Engagement:** Clear and consistent communication is essential to help users understand the benefits of the ACE and how it will impact their work. Regular updates, workshops, and Q&A sessions can help address concerns and build trust. Engaging users early in the process and involving them in decision-making can also foster a sense of ownership and commitment.
- **Training and Support:** Providing comprehensive training and ongoing support is crucial for ensuring that users are comfortable with the new tools and processes. This includes hands-on training sessions, user manuals, and access to a helpdesk or support team. By equipping users with the necessary skills and knowledge, the organization can facilitate a smoother transition and reduce resistance to change.
- **Phased Implementation:** Implementing the ACE in phases allows for gradual adaptation and minimizes disruption. Starting with a pilot phase in a specific department or team can help identify potential issues and gather feedback for improvement. This phased approach also allows for incremental changes, making it easier for users to adapt and embrace the new way of working.
- **Leadership and Advocacy:** Strong leadership and advocacy are essential for driving change and ensuring successful adoption of the ACE. Leaders should actively promote the benefits of the ACE, lead by example, and address any resistance or challenges that arise. By demonstrating commitment and support, leaders can inspire confidence and encourage others to follow suit.
- **Incentives and Recognition:** Recognizing and rewarding users who actively engage with the ACE and demonstrate positive behavior can motivate others to do the same. This can include incentives such as performance bonuses, public recognition, or opportunities for career advancement. By highlighting success stories and celebrating achievements, the organization can reinforce the desired behavior and mindset.
- **Continuous Improvement:** The implementation of the ACE should be viewed as an ongoing process rather than a one-time event. Regular feedback loops, performance monitoring, and continuous improvement initiatives can help identify areas for enhancement and ensure that the ACE remains effective and relevant. By fostering a culture of continuous improvement, the organization can adapt to changing needs and maintain the momentum of the digital transformation.

By adopting these change management strategies, ADNOC can ensure that the ACE is successfully integrated into the organization and that users fully engage with the new tools and processes. This will enable the organization to realize the full benefits of the ACE, including enhanced collaboration, improved decision-making, and increased operational efficiency.

ACE Responsibilities

Production Operations Manager

- Responsible for leading the ACE
- Oversee daily operations and ensure that production targets are met while maintaining efficiency and safety
- Encourages and develops best practices for collaboration and maintaining correct staffing in ACE
- Responsible for KPIs measuring and reporting
- Takes ownership of decision making in ACE to help resolve issues promptly and escalates as needed

Production Engineer

- Collaborate with ACE team and any other needed support to troubleshoot issues ensuring that production runs at maximum efficiency

Maintenance Planning Engineer

- Develop and coordinates maintenance plans and schedules for all equipment and systems to ensure optimal performance and minimal downtime
- Works closely with ACE team, logistics team and any other support as needed to align maintenance activities with operational goals

Maintenance Supervisor

- Acts as a focal point in ACE to coordinate the activities of maintenance teams, both with on-site and remote personnel.
- Ensures the availability and reliability of all equipment and assets through effective maintenance strategies
- Continuously improves maintenance processes to reduce downtime and costs

Reservoir Engineer

- Manages and optimizes the subsurface resources, ensuring that production strategies align with medium/long-term reservoir health and recovery goals in collaboration with ACE team
- Provides regular updates on reservoir performance and potential issues

Production Optimization Engineer

- Works closely with other ACE team members and on-site teams to monitor and analyze production data, identify issues and implement production optimization strategies
- Identifies and suggests technologies and methods to enhance production processes

Logistics Coordinator

- Acts as focal point in ACE for logistics of supplies, equipment, and personnel to ensure that all necessary resources are available to support continuous production and ensure minimal downtime
- Addresses and resolves logistics issues promptly to avoid operational disruptions
- Identifies improvement areas to enable site team to get real time visibility and address logistics issues closely liaising with maintenance planner

Data Scientist/ Engineer

- Supports decision-making processes in ACE by analyzing requested data and providing insights
- Continuously improves predictive models and algorithms to enhance potential issue identification and prescribed corrective actions

Next Generation Engineering Talent

- Participates in comprehensive training programs to learn about new technologies, ways of working and best practices.

- Works under the guidance of experienced engineers and managers to gain practical knowledge and skills, as part of ADNOC future learning development program
- Contributes fresh ideas and perspectives to improve processes and technologies

Way Forward

Conceptual designs for both Optimized and Minimized physical ACEs have been created. The recommendations for a physical ACE are based on comprehensive insights gathered from interviews and workshops, multiple rounds of discussions reviewing the outcome of the study, as well as our experience in similar work for ACE. These insights highlighted the benefits of a collaborative physical space in enhancing teamwork, innovation, and operational efficiency.

However, as per ADNOC management's decision, the way forward is to pivot towards a virtual ACE or a hybrid model. This strategic pivot reflects an understanding of modern workplace dynamics, where flexibility and technological integration are paramount for success. It also aligns with a growing trend in many organizations to embrace digital solutions that facilitate remote collaboration. The shift towards a virtual ACE acknowledges the need for a more adaptable and resilient work environment that can support remote and distributed teams.

It is important to mention here that many of the suggestions and groundwork laid for the physical ACE will still be valid and can be effectively adapted to a virtual environment. This includes leveraging digital collaboration tools, virtual meeting platforms, and other technologies to foster cohesive and productive work experience. The ACE Digital/AI toolkit, which includes various applications designed to enhance collaboration and decision-making, will be instrumental in this transition. Most of these applications will be useful for a virtual ACE, except for those pertaining to Smart Infrastructure:

- **Display Content System:** This involves high-quality large AV screens appropriate for physical workspaces with meeting/discussion rooms. While these screens are beneficial in a physical setup, their relevance may be limited in a virtual environment.
- **Smart Lighting Controls:** The IoT-based interactive lighting system was proposed for meeting rooms or focus rooms, which may be relevant only in a physical setup. These controls enhance the physical workspace but may not be necessary in a virtual setting.

While efficient designs of physical ACEs have been created, only a very high-level concept of a virtual ACE has been articulated. Therefore, the first step forward should be to continue the investigation of virtual ACE design with an accelerated process to create and test Proof of Concept (POC). This POC will help validate the feasibility and effectiveness of the virtual ACE, allowing for iterative improvements and refinements based on real-world feedback.

The transition to a virtual ACE offers several potential benefits, including increased flexibility, reduced operational costs, and the ability to support a more diverse and distributed workforce. By embracing digital transformation and leveraging advanced technologies, ADNOC can create a dynamic and resilient work environment that drives innovation, enhances productivity, and supports long-term strategic objectives.

Conclusion

In conclusion, the AI-enabled Collaborative Work Environment (ACE) at ADNOC represents a groundbreaking leap towards autonomous operations and optimized resource utilization. By seamlessly integrating people, processes, and technology, ACE revolutionizes decision-making, operational efficiency, and innovation. The project's emphasis on AI and real-time data analytics ensures that ACE remains adaptable and future-ready, aligning perfectly with ADNOC's strategic objectives.

The transition to a virtual or hybrid ACE model underscores the necessity for flexibility and resilience in today's dynamic operational landscape. The successful implementation of ACE will not only create an intelligent operational environment but also maximize value for all stakeholders, propelling ADNOC towards its vision of a fully integrated and efficient work environment. By embracing digital transformation and leveraging cutting-edge technologies, ADNOC is poised to achieve its strategic goals and maintain a competitive edge in the industry.

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